主修---Programming Part 01 (12-18)

Name: _____ ( ) Class: _____

2012

Consider the following algorithm.

```
A ← 1
B ← 1
While B < 20 do
      A ← A + B
      B ← A - B
      Output B
```

(a) Dry run the algorithm and write down the first three outputs.

_____

(2 marks)

(b) (i) The algorithm is rewritten so that the number of outputs of `B` is stored in `N`. Complete the algorithm below.

```
A ← 1
B ← 1
N ← [          ]
While B < 20 do
   A ← A + B
   B ← A - B
   N ← [          ]
   Output B
Output N
```

(ii) Write down the final values of `B` and `N`.

`B` = _____ `N` = _____

(4 marks)

(c) Suppose the initial values of `A` and `B` are changed. The values of `A` and `B` after '`Output B`' is executed for the first time are 9 and 4 respectively. What are the initial values of `A` and `B`?

`A` = _____ `B` = _____

(2 marks)

(d) The algorithm is implemented in a computer in which all memory addresses and data are 8-bit. The statement `A ← A + B` is represented by the assembly instruction

```
ADD 1000 0000, 1000 0001
```

where the contents of `A` and `B` are stored in the memory addresses `1000 0000` and `1000 0001` respectively.

Suppose that the initial values in the memory addresses are:

Memory address	Value	Decimal Value
1000 0000	0000 1011	11
1000 0001	1000 0011	131

(i) Suppose that the instruction is executed once. Write down the value stored in the following:

(1) Memory address `1000 0000` _____

(2) Memory address register (MAR) _____

(3) Memory data register (MDR) _____

(ii) How many different values can be represented by the memory addresses in the computer?

_____

(4 marks)

(a) 1, 2, 3 (①) — 2

(b) (i) 0 — 1
N + 1 — 1

(ii) B = 21 — 1
N = 7 — 1

(c) A = 4 — 1
B = 5 — 1

(d) (i) (1) 1000 1110 — 1
(2) 1000 0000 — 1
(3) 1000 1110 — 1

(ii) 256 (2^8) — 1

2013

A department store gives its members a bonus card. In a month, a member, Mary, has 10 transactions with the bonus card and the amounts are stored in `T[1], T[2], ..., T[10]`.

(a) Complete the following algorithm, `ALG1`, to calculate the total amount, `TOTAL`, that Mary has spent.

ALG1

```
TOTAL ← [          ]
I ← 1
While I [          ] 10 do
    TOTAL ← [                ]
    I ← I + 1
```

(3 marks)

The department store has a sales promotion. Every month, members do not have to pay the transaction with the lowest amount among the first 10 transactions in which the bonus card was used.

(b) (i) Complete the following algorithm, `ALG2`, to find the transaction with the lowest amount for Mary.

ALG2

```
M ← 1
J ← 1
While J <= 10 do
    If T[M] [          ] T[J] do
        M ← J
    J ← J + 1
```

(ii) Could the second line of `ALG2` be changed to '`J ← 2`'? Explain briefly.

_____

_____

(2 marks)

(c) In this sales promotion, `ALG3` is used to calculate the actual payment, `PAYMENT`, for Mary. `ALG3` includes `ALG1` and `ALG2`. Complete `ALG3`.

ALG3

Execute _____

Execute _____

PAYMENT ← _____

(3 marks)

(d) The department store has another sales promotion. The 20 members with the highest total spending each month can enjoy 20% off all year round. A computer application system is used to find these 20 members.

(i) Should the system be a single-user system or a multi-user system? Explain briefly.

(ii) Should the system be a batch processing system or a real-time system? Explain briefly.

(4 marks)

(a) 0 1
<= 1
TOTAL + T[I] 1

(b) (i) >= 1

(ii) Yes, it is because the first comparison in the while loop (T[1] >= T[1]) is redundant. 1

(c) Execute ALG1 (or ALG2) 1
Execute ALG2 (or ALG1) 1
PAYMENT ← TOTAL - T[M] 1

(d) (i) Single-user system: As the system is used to find the 20 members, one single user should be sufficient to generate the list of the members. 1 + 1

(ii) Batch processing system: As the system will generate the list based on the monthly total amounts, it should be a batch processing system to perform the tasks in one batch each month. 1 + 1

2014

The following algorithms ALG1 and ALG2 help George find a sequence of numbers.

ALG1	ALG2
A ← 0 B ← 1 P ← 10 While A <= P do Output A temp ← A + B A ← B B ← temp	A ← 0 B ← 1 Q ← 5 For I from 1 to Q Output A temp ← A + B A ← B B ← temp

(a) (i) Write down the output values of ALG1.

(ii) Write down the output values of ALG2.

(iii) If the output values of ALG2 need to be the same as those of ALG1, what should the initial value of Q be changed to?

(5 marks)

George further develops the algorithms to build an Internet application.

(b) (i) If the application is a parallel processing system, how does the application work?

(ii) If the application is a distributed processing system, how does the application work?

(4 marks)

(c) George can use either freeware or open source software to develop the application.

(i) What is the major common characteristic of the two types of the software?

_____

_____

(ii) What is the major difference in terms of the resources that the software provides?

_____

_____

(iii) Why do some people promote open source software instead of freeware?

_____

_____

_____

(4 marks)

(a) (i) 0, 1, 1, 2, 3, 5, 8 (-1 for 1 mistake) 2

(ii) 0, 1, 1, 2, 3 (-1 for 1 mistake) 2

(iii) 7 1

(b) (i) When the application is executed, a process is divided into different portions/computational threads and executed in different CPUs independently at the same time. 1, 1

(ii) The system consists of components which interact with each other (managed by a program) through a networked environment in order to execute the application. 1, 1

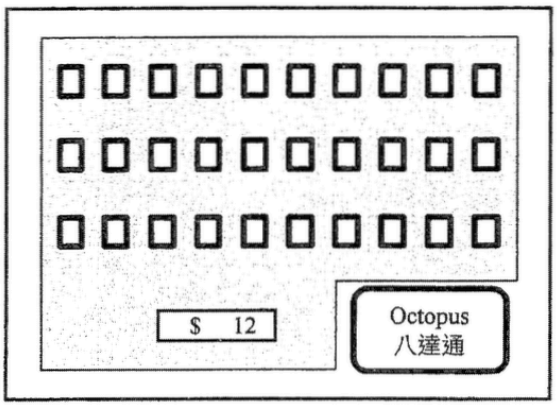
(c) (i) There is no purchase cost for non-commercial purpose. 1

(ii) Source codes are given in open source software. 1

(iii) Developers and users (multiple parties) can contribute and develop the software continuously. 1,1

2015

Ms Wong develops a vending machine which has an embedded computer. She writes a program to control the sales of 30 kinds of products in the vending machine.



(a) (i) Other than the operating system, what will the secondary storage in the embedded computer store?

_____

_____

(ii) The operating system is not installed in the ROM of the embedded computer. Why not?

_____

_____

(iii) The vending machine only allows customers to use the Octopus stored value card to buy the products. What is the advantage and disadvantage of this setting?

Advantage: _____

_____

Disadvantage: _____

_____

(4 marks)

Ms Wong uses an array with 30 elements, `P[1]`, `P[2]`, ..., `P[30]`, to store the numbers of the 30 kinds of products remained in the vending machine. She uses the following algorithm ALG1 to check whether all the products in the machine are sold out or not.

```
ALG1
N ← 0
For I from 1 to 30
    N ← N + P[I]
If N = 0 then
    FLAG ← FALSE
else
    FLAG ← TRUE
```

(b) (i) If the values of `P[1]` and `P[2]` are `0` and `1` respectively, what is the final value of `FLAG`?

_____

(ii) What does the final value of `N` mean?

_____

(iii) Ms Wong chooses Boolean as the data type of `FLAG`. What is the major benefit of her choice?

_____

(4 marks)

(c) (i) According to the values of `I` and `FLAG`, complete the following truth table.

I	FLAG	(I <= 30) AND (NOT FLAG)
1	FALSE	TRUE
15	TRUE	
30	FALSE	
31	TRUE	

(ii) The following algorithm ALG2 performs the same task as ALG1.

```
ALG2
FLAG ← FALSE
I ← 1
While (I <= 30) AND (NOT FLAG) do
    IF P[I] > 0 THEN
        FLAG ← TRUE
    I ← I + 1
```

Which algorithm, ALG1 or ALG2, should Ms Wong use? Explain briefly.

_____

_____

(4 marks)

(a) (i) It stores the program that monitors the sales of the products. / Drivers 1

(ii) It is not easy to update the operating system. 1

(iii) Advantage:
Individual customers' behaviour on using the vending machine can be recorded. /
Reduce the risk of cash handling / fast transaction / no change is required. 1
Disadvantage:
One single payment method may limit the sale. 1

(b) (i) TRUE 1

(ii) the total number of items remained 2

(iii) It has only two possible values so that the program will be more readable /
the control of data is more strict. 1

(c) (i) 1×3

I	FLAG	(I <= 30) AND (NOT FLAG)
1	FALSE	TRUE
15	TRUE	**FALSE**
30	FALSE	**TRUE**
31	TRUE	**FALSE**

(ii) ALG2 – It generates a fewer number of iterations / average execution time is shorter. 1
ALG1 – It may have fewer conditional statements.

2016

One day, Charles receives a spam mail below:

> Hi,
>
> Congratulations! You have won a lottery. Please click on the following URL and log in to your bank account to collect your money.
>
> http://lottery.ibank.pl
>
> Mr Peter Chan
> Customer Service Manager

(b) If Charles clicks on the URL, what will the DNS do subsequently?

(1 mark)

(c) Although Charles clicks on the URL, he does not input his bank account information. Give **two** potential threats that he might encounter.

(2 marks)

The email server can identify spam mail by matching strings in a data file. The array `ST` stores the strings in the data file, as shown below:

```
ST[1] ← "CONGRATULATIONS"
ST[2] ← "LOTTERY"
......
ST[1000] ← "WINNER"
......
```

The following two algorithms, ALG1 and ALG2, check whether the string stored in the variable `check_ST` is included in the data file. They produce the same result. `N` is a variable for storing the total number of strings in the data file.

ALG1	ALG2
Input check_ST FOUND ← -1 For I from 1 to N If check_ST = ST[I] then FOUND ← I	Input check_ST FOUND ← -1 I ← 1 While (I<=N) and (FOUND = -1) If check_ST = ST[I] then FOUND ← I I ← I + 1

(d) (i) Suppose that the string in `check_ST` is `"LOTTERY"`. After executing the algorithms, what is the value of `FOUND`?

_____

(ii) Give **two** test values of `check_ST` to verify the algorithms effectively and describe your answer briefly.

_____

_____

(iii) In the case of (d)(i), will ALG1 or ALG2 be more efficient? Explain briefly.

_____

_____

_____

(iv) What is the benefit of using an array to store the strings?

_____

_____

(6 marks)

(b) Translate the domain name into its IP address. 1

(c) Malicious code might be spread onto Charles' computer. / Charles' computer will be attacked through program bugs of browser. (virus infection) 1×2
Spammers will know that Charles responds to the spam mail and then send him more spam mails. (spam mail)
It links to web sites that Charles does not intend to visit. (phishing)

(d) (i) 2 1

(ii) `LOTTERY, HELLO` 1
They are one correct string and one incorrect string to validate the algorithms. 1

(iii) ALG2 is more efficient as it has not gone through all strings in `ST`. 2*

(iv) It is easy to implement a loop to check all strings. 1

2017

某鐵路公司開發一個電腦系統，在車站設置具有觸式顯示屏幕的資訊亭。

(a) 資訊亭的顯示屏幕上應顯示以下列車的資訊，並協助乘客查找列車的到達時間和月台編號。

- 列車編號，如 3338 和 0910
- 目的地，如深圳和天津
- 到達時間，如 14:30 和 08:00
- 月台編號，如 3 和 12

試設計資訊亭的互動式版面，並加注釋。

(3 分)

(b) 資訊亭上的數據儲存在可電刪可編程唯讀記憶體（EEPROM），它是快閃記憶體的一種。

(i) 舉出快閃記憶體與 RAM 的一個不同之處。

_____

_____

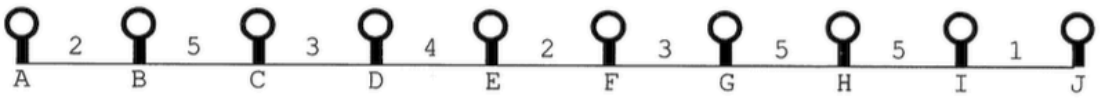
(ii) 舉出快閃記憶體與 ROM 的一個不同之處。

_____

_____

(2 分)

在某一條路線上，從車站 A 到車站 J 共有 10 個車站，任何兩個相鄰車站之間均有一個估計的行車時間（以小時計），如下所示。舉例來說，從車站 A 到車站 B 的行車時間估計為 2 小時。



(c) 從車站 D 到車站 G 的行車時間估計是多少？ _____

(1 分)

在此電腦系統中，使用了陣列 W 中的 W[1],W[2],…, W[9] 來分別儲存 9 個估計的行車時間，即 W[1]=2 ， W[2]=5 ，如此類推。

(d) 完成以下算法，以計算從車站 A 到車站 J 估計的行車時間。

X ← _____

Y ← _____

T ← 0

設 i 由 X 至 Y

　　T ← _____ + _____

輸出 T

(4 分)

(e) 在循環內使用變量 X 和 Y 而不使用常數的主要優點是什麼？

_____

_____

_____

(2 分)

(a) ① 顯示所有信息 3
① 互動版面
① 易於輸入

(b) (i) 快閃記憶體（EEPROM）是非易失性的。 1

(ii) 快閃記憶體（EEPROM）是可重寫的。 1

(c) 9 1

(d) X ← 1 1
Y ← 9 1
T ← T + W[i] 1, 1

(e) 靈活替換在算法中各個車站的索引， 1
以估計路線中不同行車段的行車時間。 1

2018

Mary designs an algorithm of a number guessing game as follows:

```
N ← 100
ANSWER ← a random integer between 1 and N inclusive
do
    input an integer into GUESS
    if (GUESS < 1) or (GUESS > N) then
            output  "Out of range!"
    else if (GUESS <> ANSWER) then
            output  "Try again!"
until (GUESS = ANSWER)
output  "You win!"
```

(a) Suppose the value of ANSWER is 19. Suggest an input value to test the algorithm for each output.

Input	Output
	Out of range!
	Try again!
	You win!

(3 marks)

(b) Mary plans to make two modifications to the algorithm:

1. If GUESS is smaller than ANSWER, display "Too small!" and if GUESS is larger than ANSWER, display "Too large!";
2. Allow users to guess at most 5 times.

(i) Complete the algorithm below showing the modifications.

Line number	Content
1	K ← 0
2	N ← 100
3	ANSWER ← a random integer between 1 and N inclusive
4	do
5	input an integer into GUESS
6	if (GUESS < 1) or (GUESS > N) then
7	output "Out of range!"
8	else if([ ]) then
9	output "Too small!"
10	else if([ ]) then
11	output "Too large!"
12	K ← [ ]
13	until (GUESS = ANSWER) or ([ ])
14	if (GUESS = ANSWER) then
15	output "You win!"

(4 marks)

(ii) If Line 14 'if (GUESS = ANSWER) then' is omitted, the output of the algorithm may be wrong. Why?

(1 mark)

(c) Mary completes the number guessing game and uploads the program to the Internet for the public to play.

(i) Mary licenses the program as open source software. What can users do with the program? Give two examples.

(2 marks)

(ii) Some users install the program on their mobile devices and play the game for a long time every day, leading to some health problems. Suggest two ways to relieve these health problems, other than taking breaks.

(2 marks)

Answers written in the margins will not be marked

(a) Any integer small than 1 or greater than 100 1
Any integer between 1 and 100 inclusive, except 19 1
19 1

(b) (i) `GUESS < ANSWER`
`GUESS > ANSWER` 1
`K + 1` 1
`K = 5` or `K > 4` or `K >= 5` 1
1

(ii) As the loop may end due to `K = 5`, which does not mean the user wins. Even if a user cannot correctly guess the answer, the message will be displayed anyway. 1

(c) (i) Other people can
- freely use the program
- change the program code to improve the game (read/modify)
- freely distribute the game

1x2

(ii) Control the brightness of the screen. (filter)
Enlarge the font size. (accessibility function)
Use voice input/output.

1x2